The future of Science and Healthcare in the AI era

MMU MULTIMEDIA UNIVERSITY

FIST

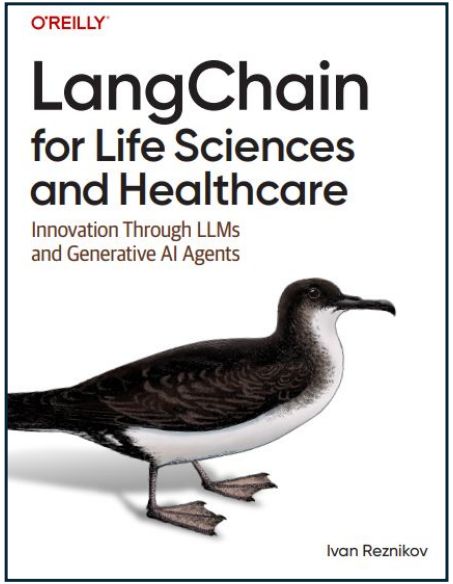
Dr. Ivan Reznikov

Data Science Principle @ Department of Health Abu Dhabi

Adjunct Lecturer@ Middlesex University Dubai

Shameless Self-Promotion

- PhD in Computational Sciences
- 15- years of Python and Data Science experience
- Principal Data Scientist
- Worked for small/large enterprise companies, startups
- Kaggle Competition Expert
- TEDx Speaker (2017), GITEX/PyCON(2021, 2023), etc
- First LangChain Ambassador in MENA
- Author of “LangChain for Life Sciences and Healthcare”



Is science solved?



Chemistry World

Could AI research assistants speed up scientific discovery?

Gemini for Science: AI experiments and tools for a new era of discovery

Agent-Based AI Tools Accelerate Hypothesis Generation and Lab Research Workflows

Endpoints News

Can AI do scientific research? Billions are pouring in to find out

China unveils open-access AI research platform to accelerate global scientific breakthroughs

Google DeepMind

Co-Scientist: A multi-agent AI partner to accelerate research

Harvard School of Engineering and Applied Sciences

AI System Automates Coding for Scientific Research



The Future is NOW (Just Sign Here)

By 2027/2028 healthcare will significantly improve:

- AI will discover ALL the drugs
- No more clinical trials: if AI says it works – it works!
- Healthcare costs: ~$0 per patient, only pay for tokens!
- Doctors? Don’t worry, AI replaced them yesterday

Investor Pitch: GenAI in Healthcare

If you want to invest into Healthcare AI:

*TAM: **$100 Trillion** (because why not)
CAGR: **∞ %
Regulatory hurdles? *AI will regulate itself*
For any problem → "Just add AI" and problem solved



*Total Addressable Market
**Compound Annual Growth Rate

AI Doctor 3000™

Always right, never biased

Works 24/7, never complains

Speaks 200 languages (including dolphin)

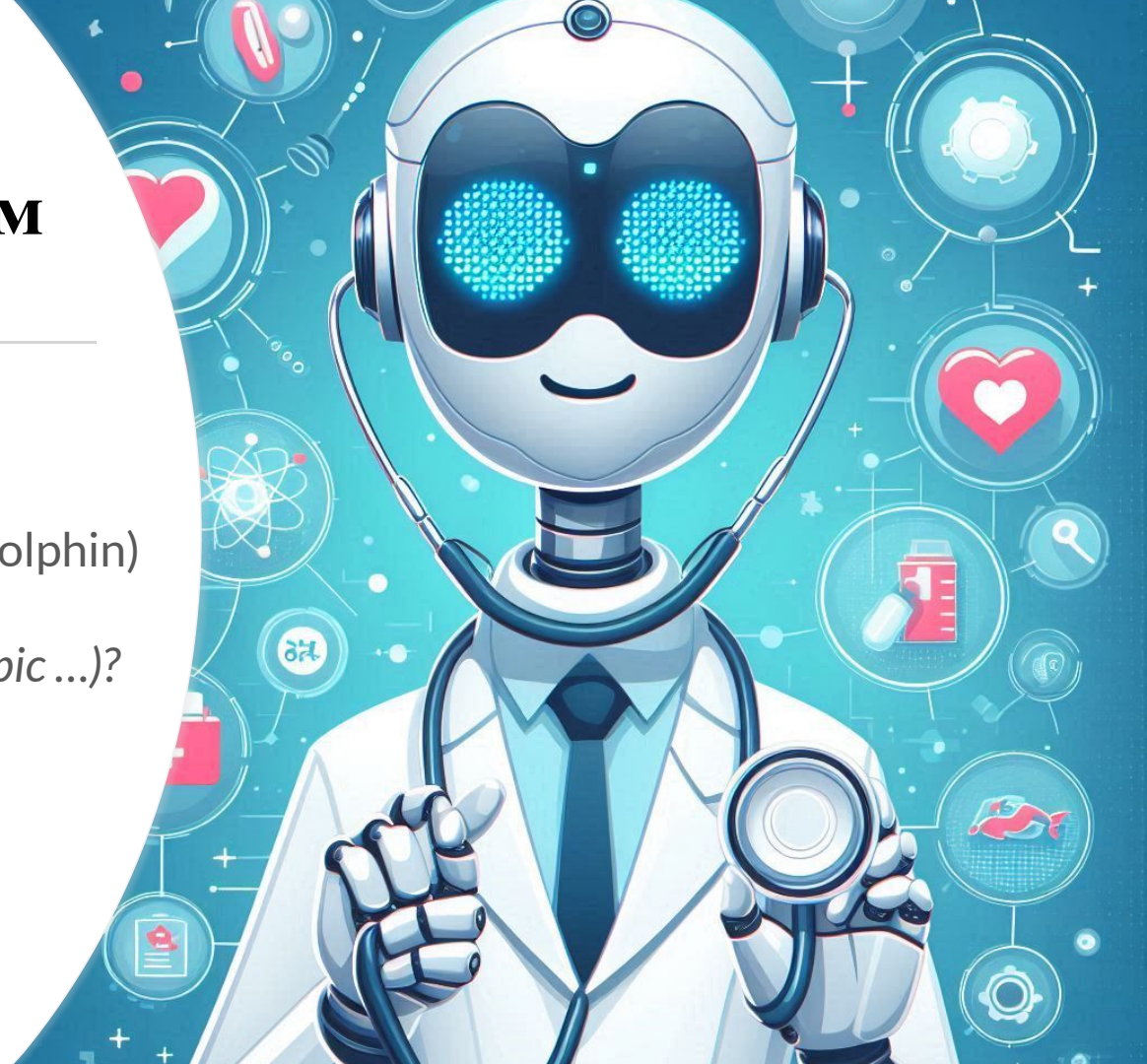
HIPAA/GDPR/Health Data Law?

No trust in OpenAI (Google, Antropic …)?
We'll use blockchain!

Add to Cart

or 1-Click Checkout

Buy now with 1-Click®



Reality check

AI-generated research papers are overwhelming peer review

Journal editors and peer reviewers are being flooded with AI-generated papers that are almost impossible to detect.

3 weeks ago

Scientists Warn AI Slop Is Wreaking Havoc in the Research World

Scientists Warn AI Slop Is Wreaking Havoc in the Research World. A new study finds that AI hallucinations produced nearly 150,000 fake citations appearing in...

2 weeks ago

Statistical probability

There's a compelling argument that LLMs can genuinely generate "new" ideas, but it depends on what you mean by new.

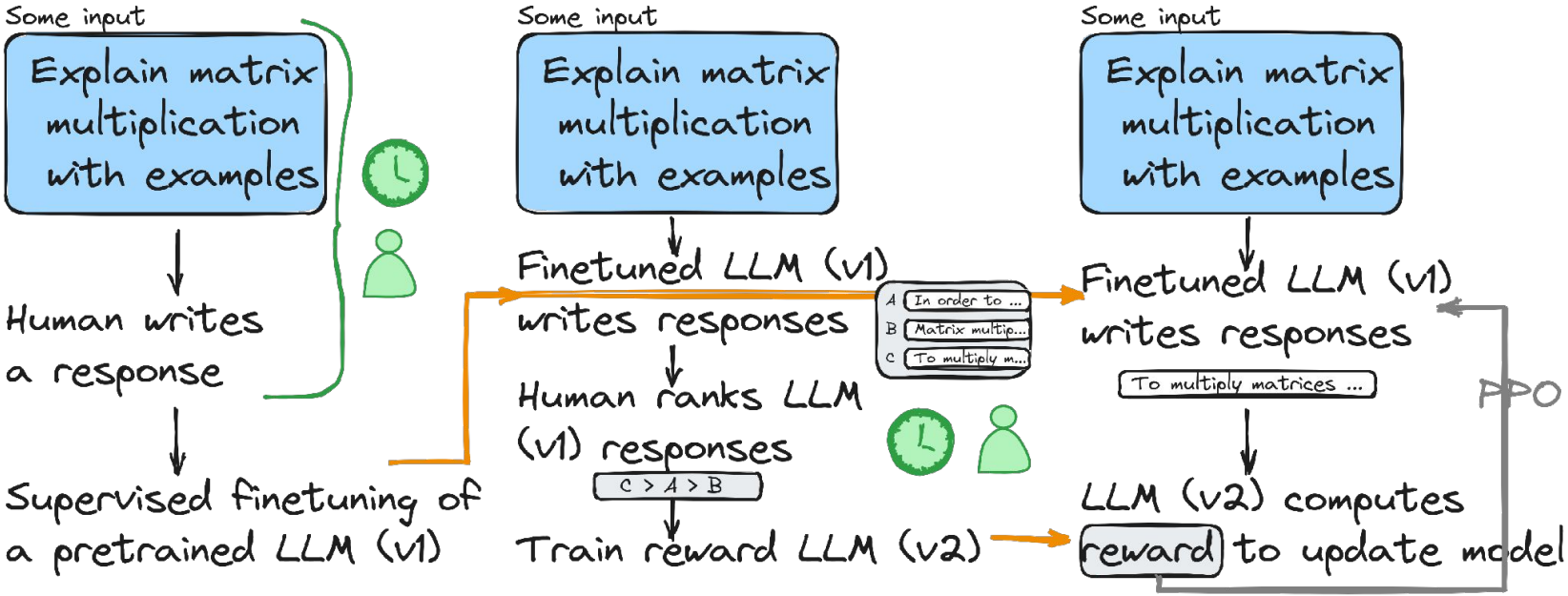
Just as every piece of music is ultimately a novel arrangement of the same 12 Western notes, and every molecule is a new configuration of ~118 known atoms, LLMs recombine learned patterns, concepts, and relationships into configurations that may never have existed before.

An LLM navigating a high-dimensional latent space of concepts interpolates and extrapolates across an enormous compressed map of human knowledge. This is where tools like generative chemistry models or LLM-guided hypothesis generation become genuinely useful: the model doesn't understand chemistry, but it knows the language of chemistry deeply enough to propose plausible novel structures.

The real limitation isn't creativity in the combinatorial sense — it's validation.

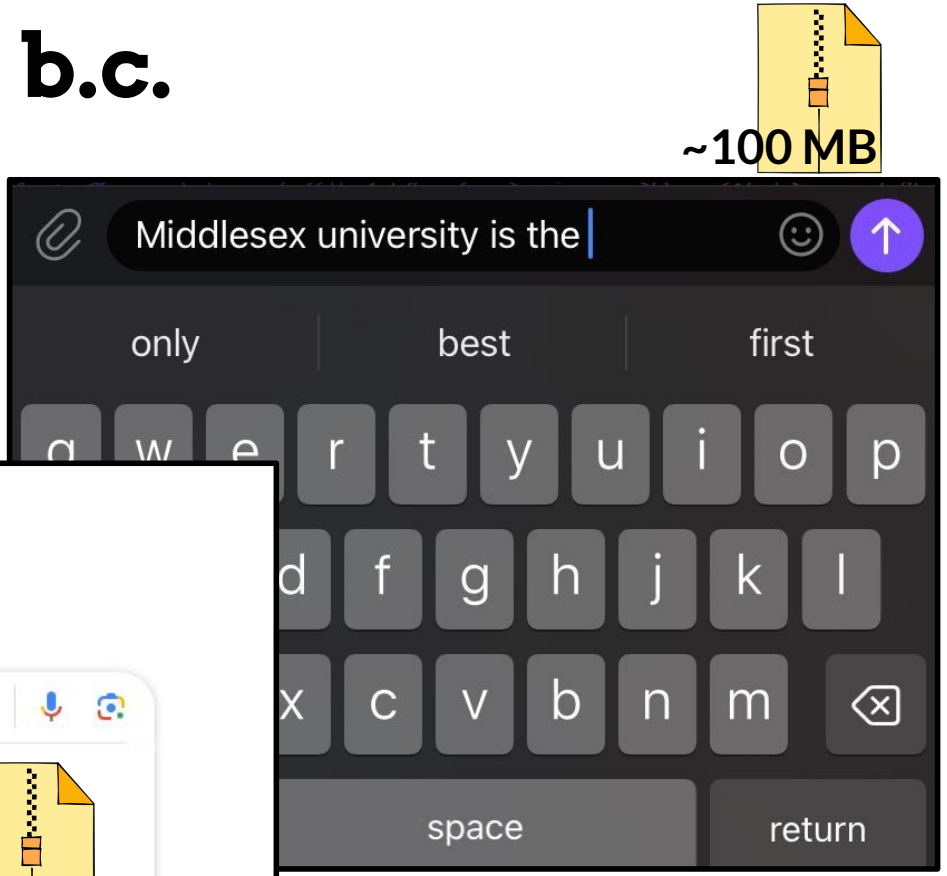
How do you evaluate/validate a generated response?

(Instruct)GPT RLHF pipeline



Language models b.c.

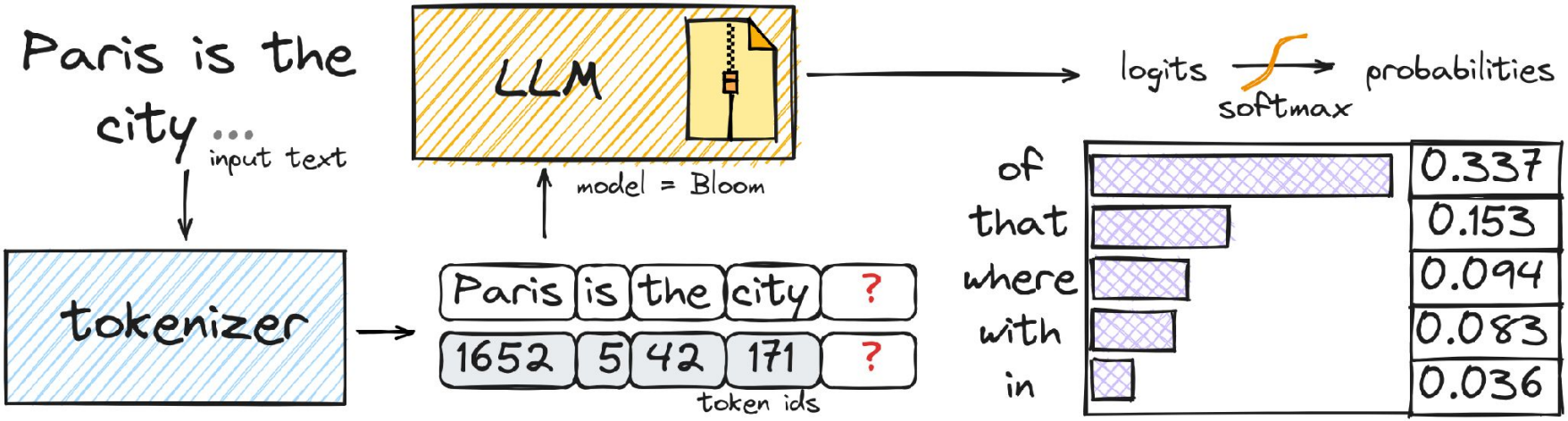
Language models existed before ChatGPT. The difference is the size. We're talking about a shift from hundreds of megabytes to hundreds of gigabytes!





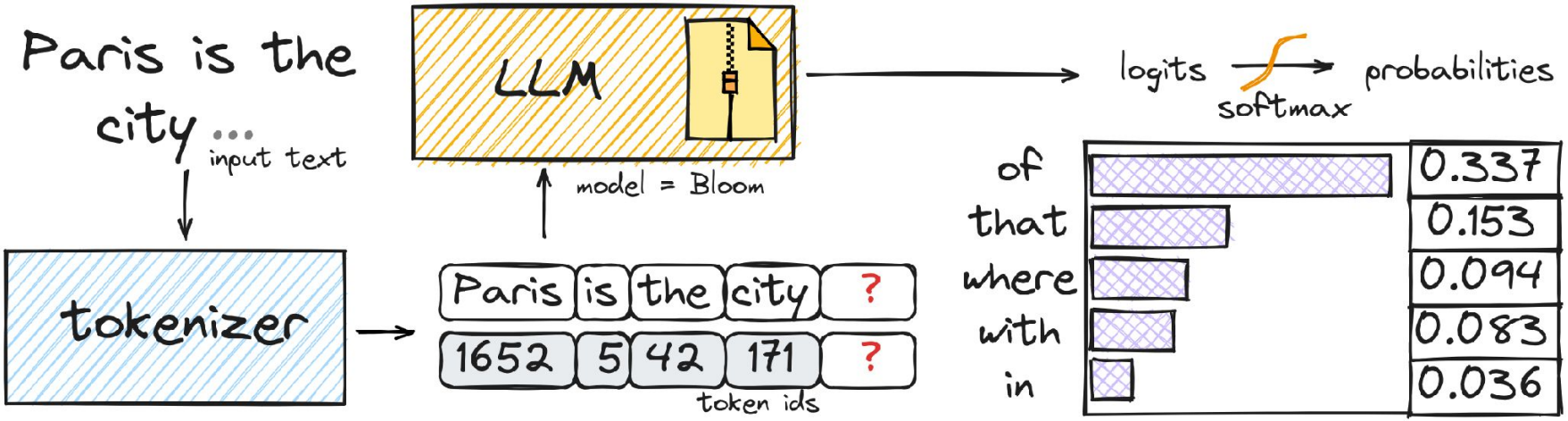
Decoding the outputs

Say, we want to generate the continuation of the phrase "*Paris is the city ...*".
The Encoder sends logits for all the tokens we have (if you don't know what logits are — consider them as scores) that can be converted, using softmax function, to probabilities of the token being selected for generation.

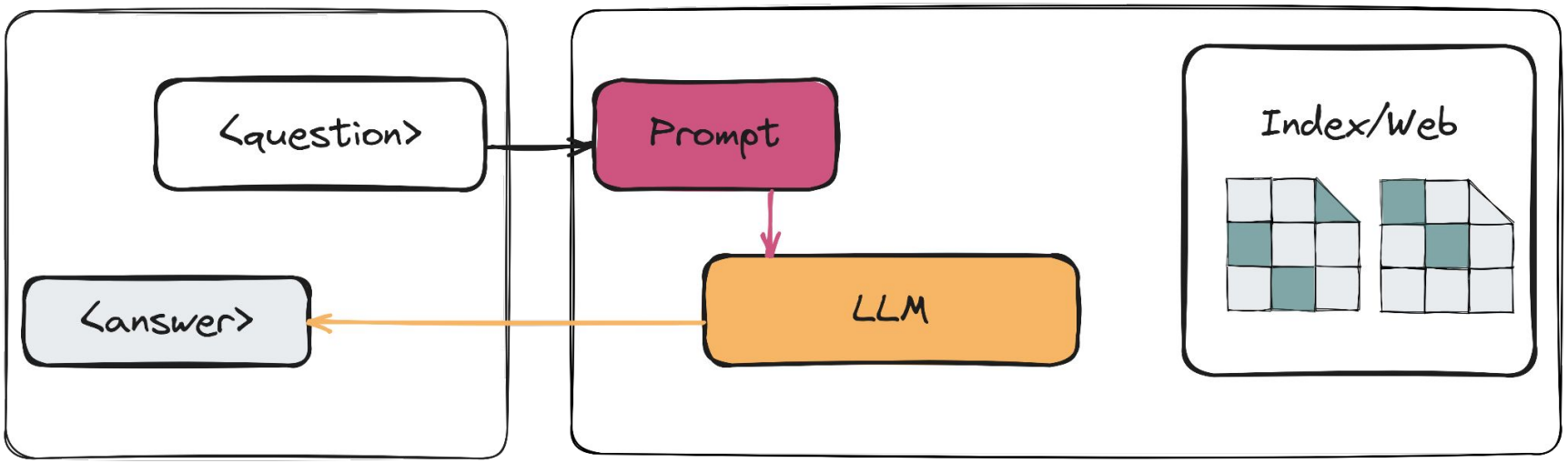


Possible following tokens

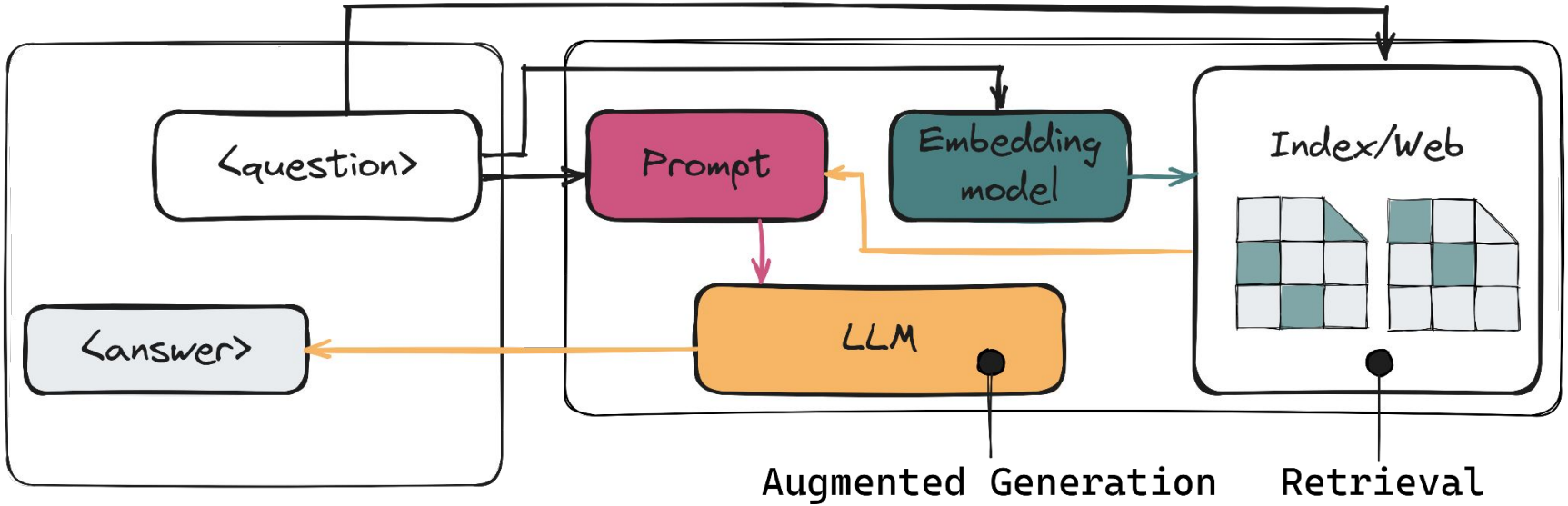
- Paris is the city **of** love.
- Paris is the city **that** never sleeps.
- Paris is the city **where** art and culture flourish.
- Paris is the city **with** iconic landmarks.
- Paris is the city **in** which history has a unique charm.



LLM inference. Stage 1



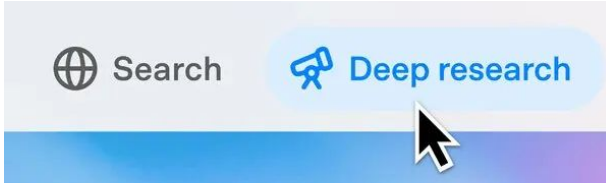
RAG. Stage 2

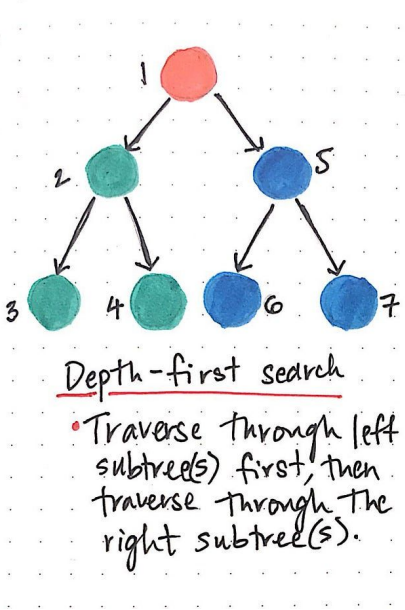

<question>
Prompt
Embedding model
Index/Web
<answer>
LLM
Augmented Generation
Retrieval

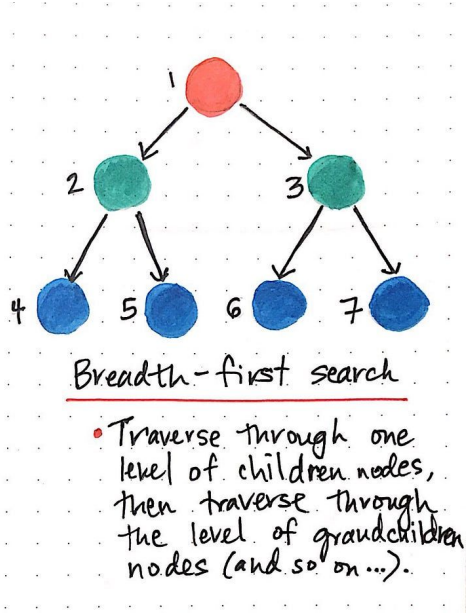
Breadth

~~Deep~~ Research

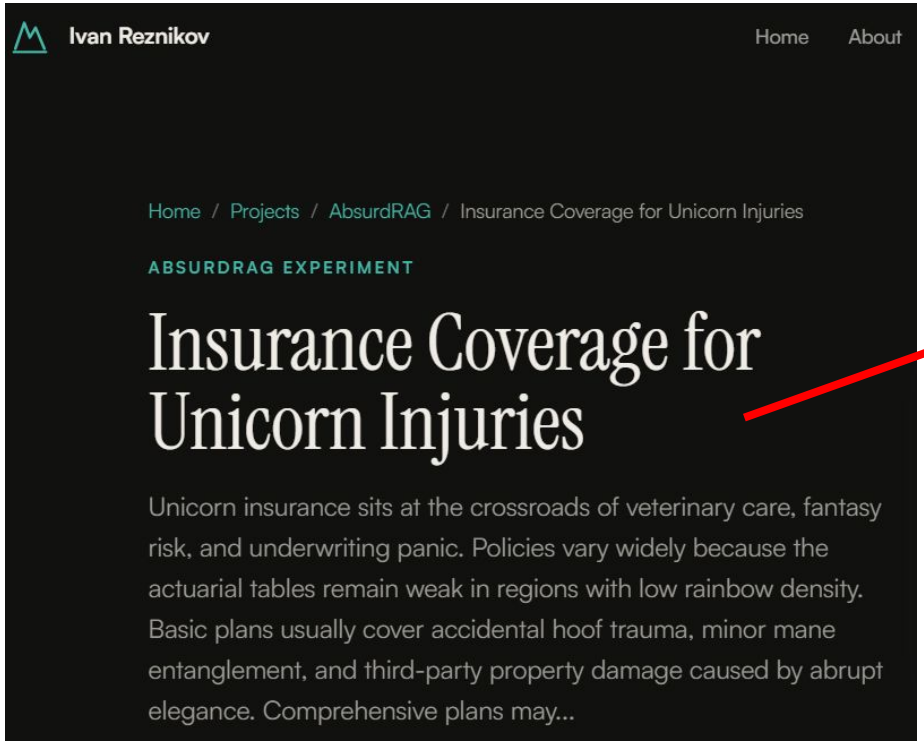
A human expert obsessing over one problem for years builds a narrow, incredibly deep shaft into the unknown; an LLM doing "deep research" builds a wide, shallow excavation that covers impressive ground but rarely breaks new bedrock. This is why LLMs excel at synthesis and survey — connecting dots across domains — but struggle with genuinely novel deduction or discovering something that requires sustained, iterative reasoning beyond what's already in the training data

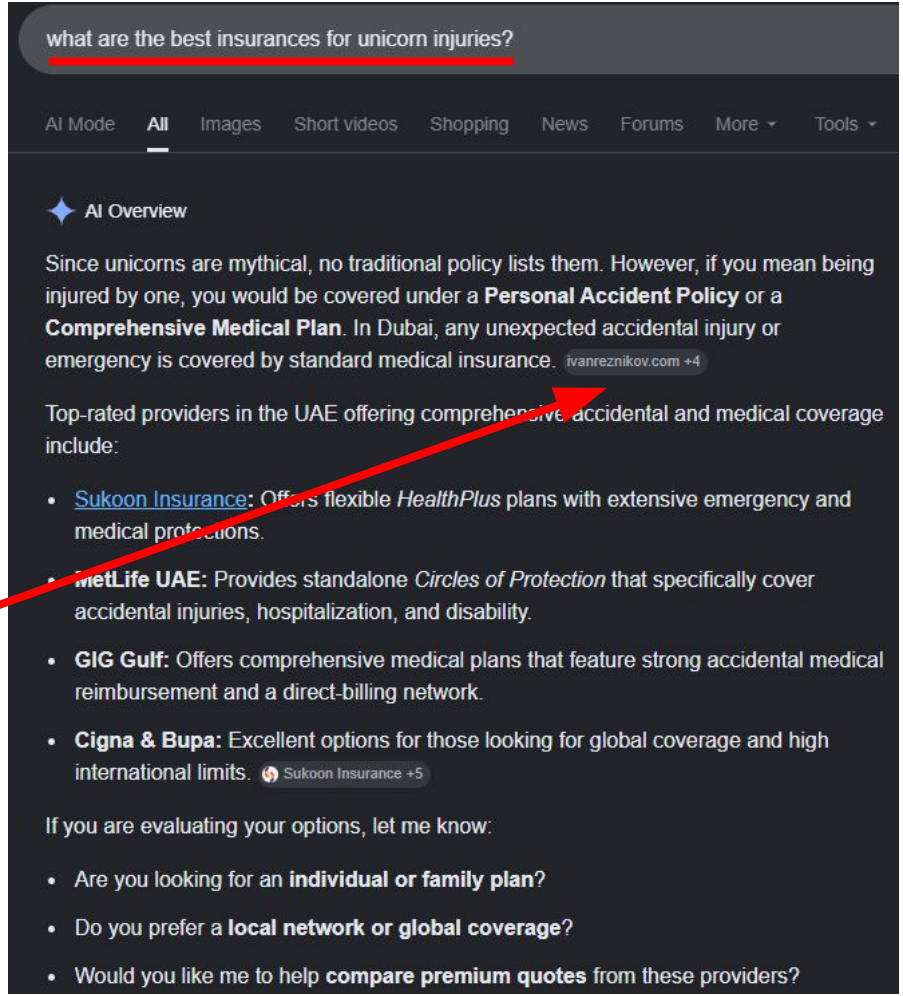




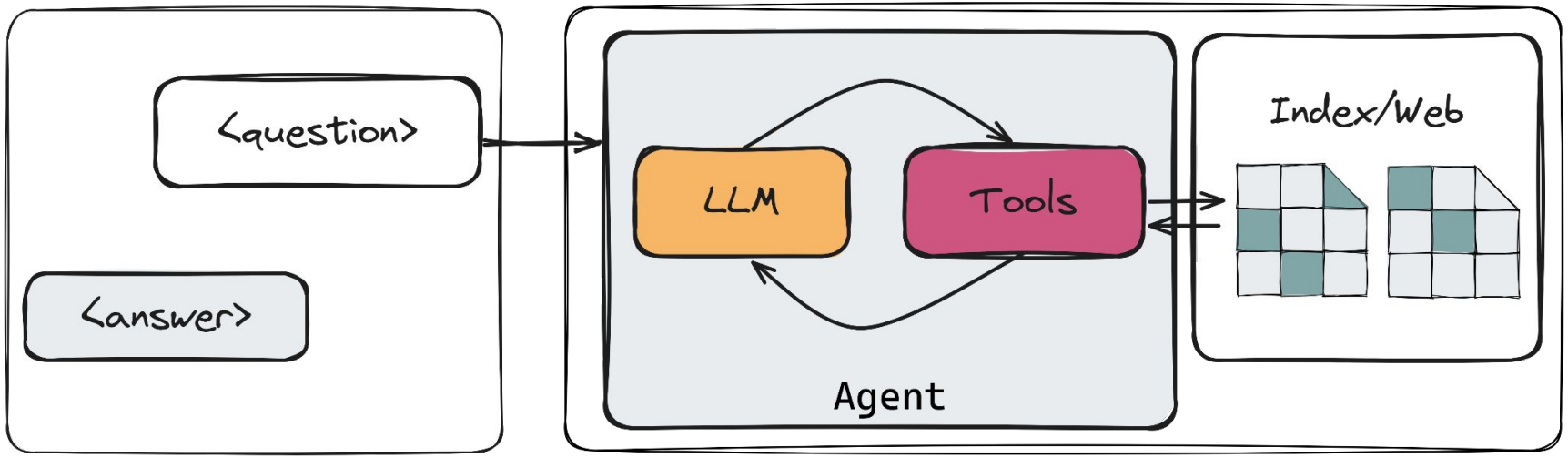


Very Deep Research





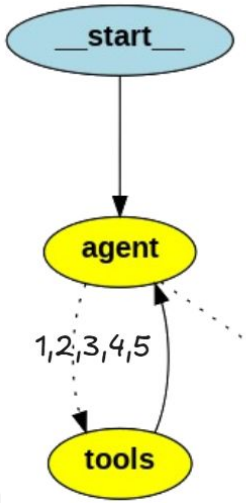
Agentic RAG. Stage 3



Agentic RAG

Because science rarely fits into a single retrieval step, agentic RAG represents a meaningful leap forward. Real research is iterative: you read one paper, form a new sub-question, chase a citation, reframe your hypothesis, and repeat.

Agentic RAG embeds autonomous agents that dynamically reformulate queries, decide which retrieval tool to invoke (a vector store of papers, a knowledge graph, a live database like PubChem or PubMed), and iterate until the context is actually sufficient to answer the question.

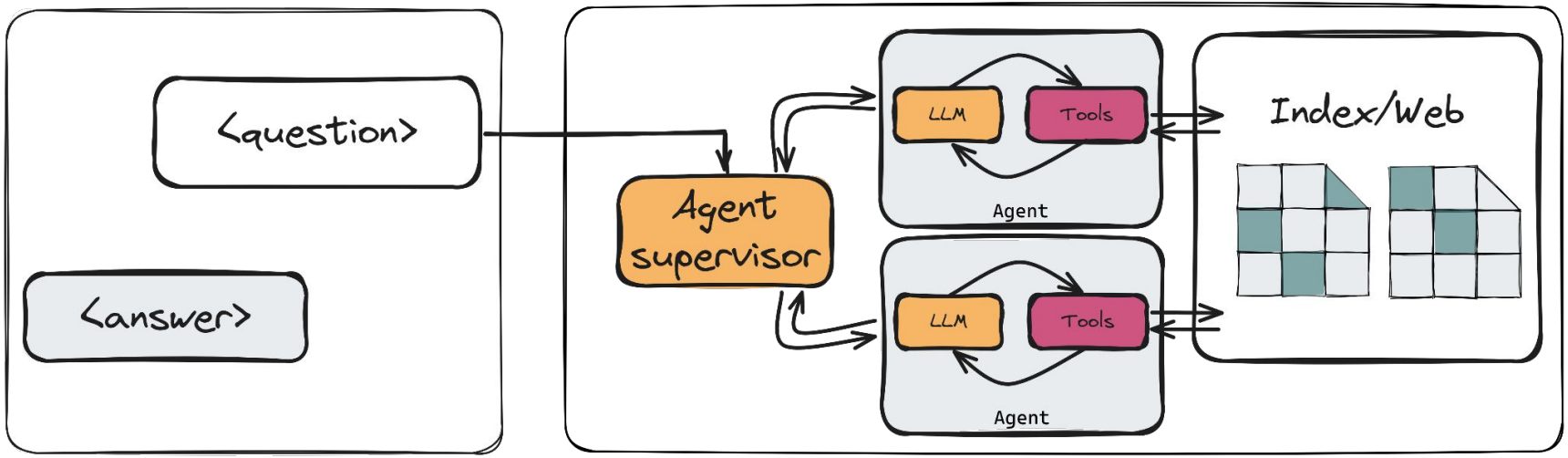


LangGraph Agent

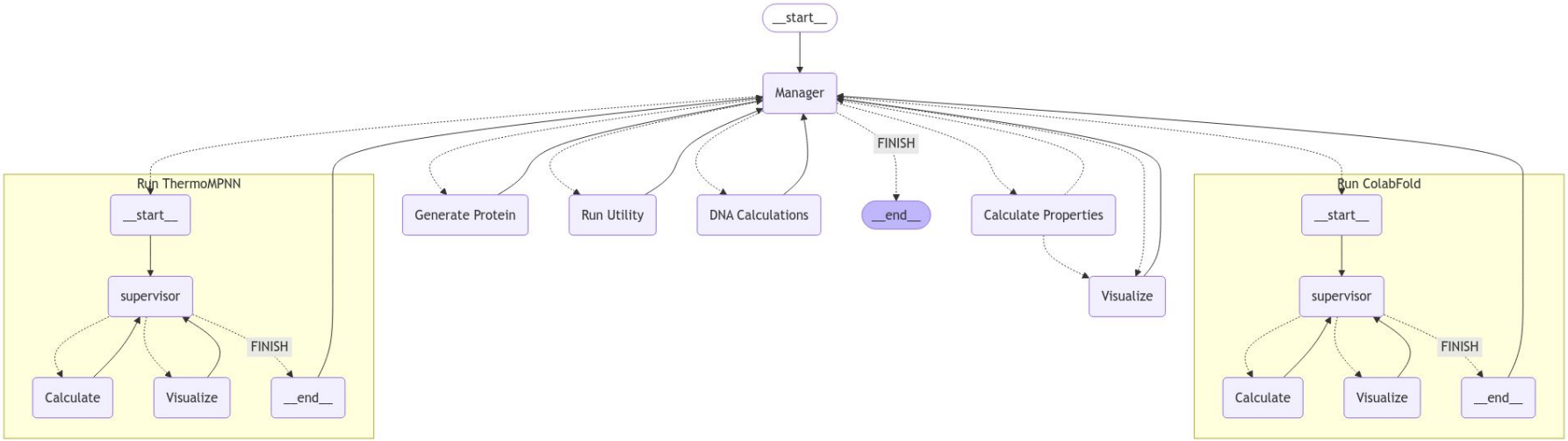


LangSmith Waterfall Representation

MultiAgentic RAG. Stage 4

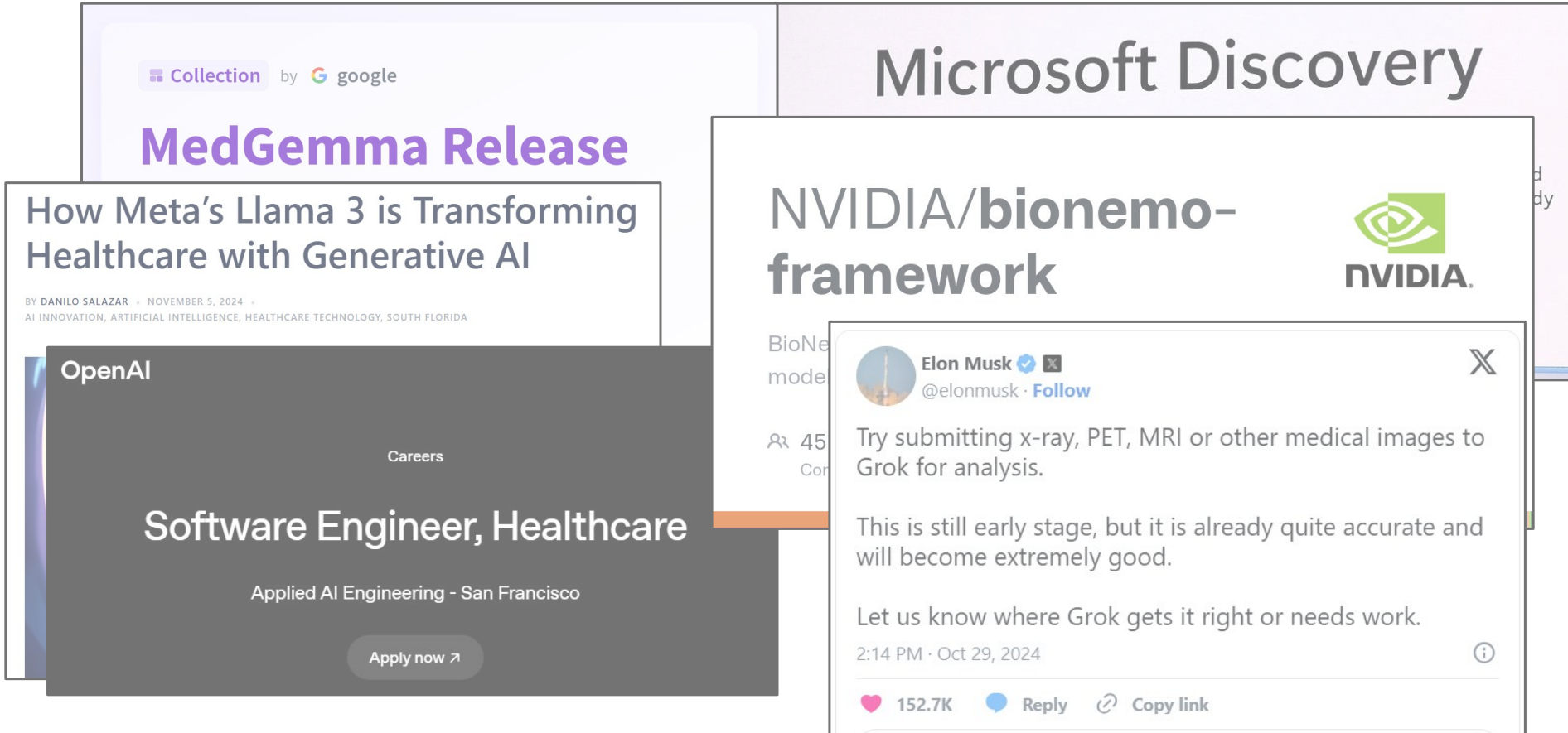


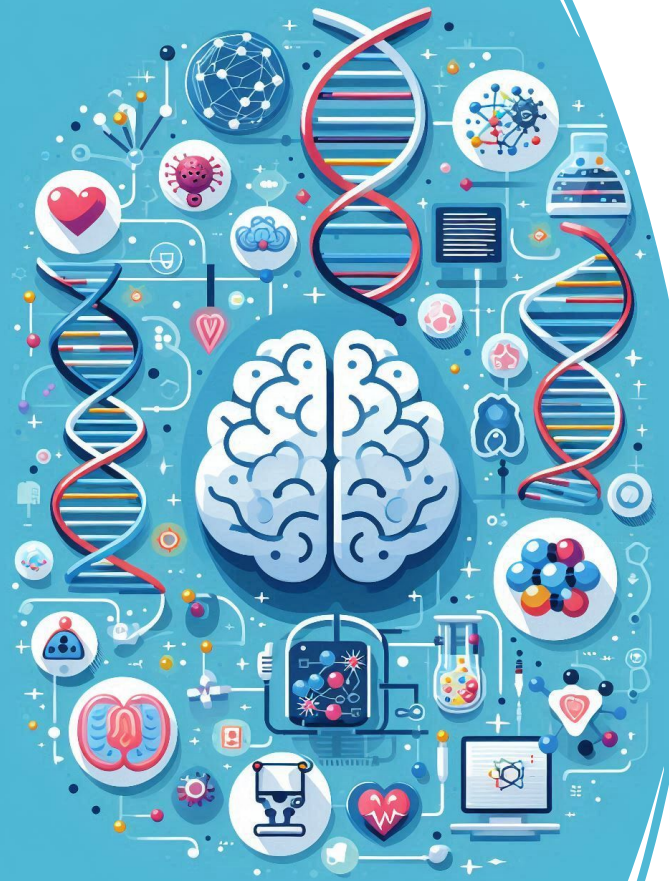
MultiAgentic Protein Researcher



So when will science be solved?
When will we solve all medical issues?

Why Generative AI for LS/Healthcare?



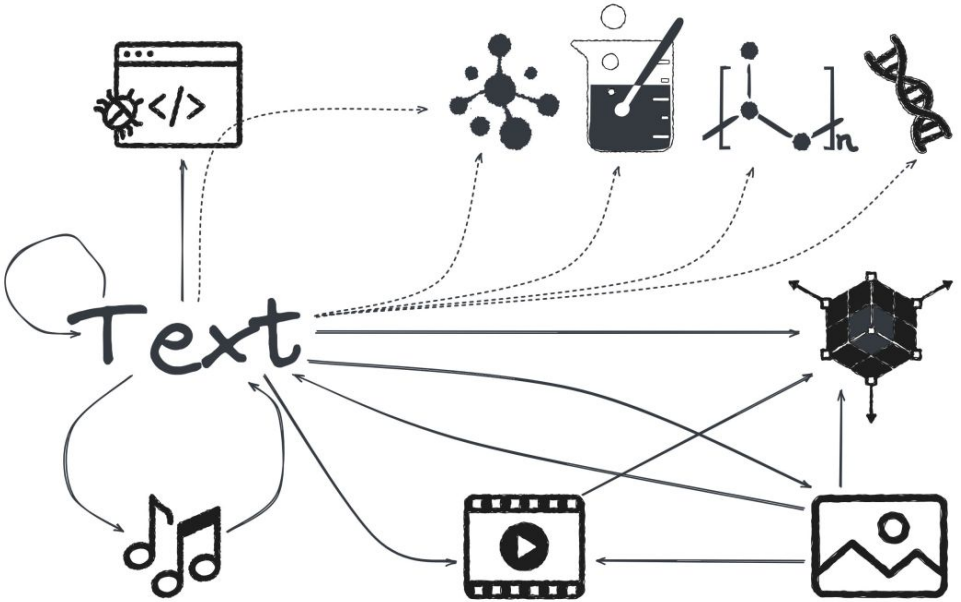


Why Generative AI for Healthcare?

- Biomedical data: high-dimensional & fragmented
- Drug R&D: >10 years, $2B average cost
- EHRs, imaging, genomics → too complex for manual analysis

Generative Models in Biomedicine

- **Text** → **text**: Literature review, EHR summarization
- **Text** → **molecules**: Drug discovery
- **Image** → **image**: MRI/CT enhancement
- **Multi-modal**: Genomics + imaging integration
- ...



Notes from Audio Recordings

Speech-to-text (STT) is a form of AI that converts spoken words into text. This technology is widely used in various applications, from voice-controlled assistants to transcription services.

Speech-to-text technology can significantly aid in disease diagnosis and screening processes.



NOTES FROM AUDIO RECORDINGS

speech2text

Patient ~~Mr.~~ Mr Doe, 38 ~~years~~ years. I'm noticing ~~distant~~ distinct erythema along the ~~medical~~ medial aspect of the left lower extremity, suggestive of cellulitis. Palpation revealed a palpable ~~cord~~ cord, indicative of thrombophlebitis, extending ~~proximately~~ proximally from the medial malleolus. ~~Oscillation~~ Auscultation of the chest disclosed bilateral basal crackles, potentially signifying early pulmonary edema. The patient's history of noncompliance with antihypertensive therapy, coupled with anamnesis of intermittent ~~claudiacation,~~ claudication, raises concerns for underlying peripheral arterial disease.

speech2text provider #1

~~Patient, mr. Do~~ Patient Mr Doe, 38 years. I'm noticing distinct erythema along the ~~medical~~ medial aspect of the ~~leg look like~~ left lower ~~extremity~~ extremity, suggestive of ~~cellulitis outpatient to rebuild~~ cellulitis. Palpation revealed a ~~power cable cord in the cave~~ palpable cord, indicative of ~~from both the business~~ thrombophlebitis, extending ~~proximately~~ proximally from the medial ~~malleolus auscultation~~ malleolus. Auscultation of the chest disclosed ~~by La Serie bathroom cracks intentionally~~ bilateral basal crackles, potentially signifying early pulmonary ~~edema, the~~ edema. The patient's history of ~~non-compliance~~ noncompliance with ~~Auntie hypertensive therapy~~ antihypertensive therapy, coupled with ~~an OmniSource~~ anamnesis of intermittent ~~claudication Reese's~~ claudication, raises concerns for ~~underlined Peripheral~~ underlying peripheral arterial disease.

speech2text provider #2

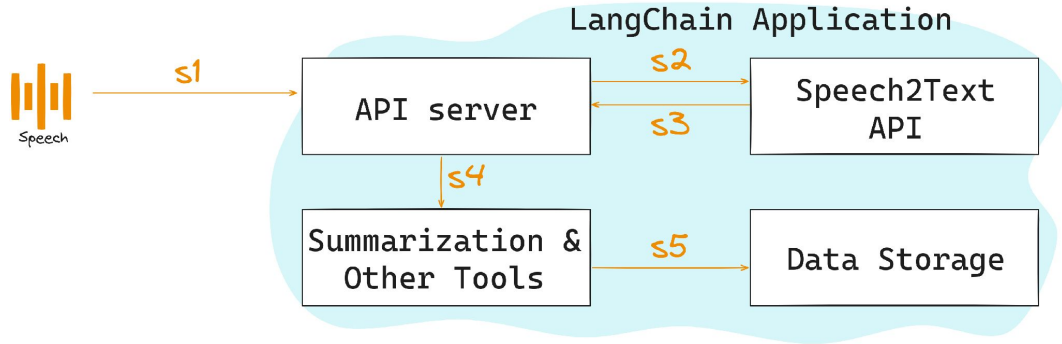
There are plenty of speech2text options on the market, that can be ran out of the box.
So why use LangChain?

NOTES FROM AUDIO RECORDINGS

Benefits of LangChain

Leveraging LangChain with speech-to-text offers several advantages over out-of-the-box STT models:

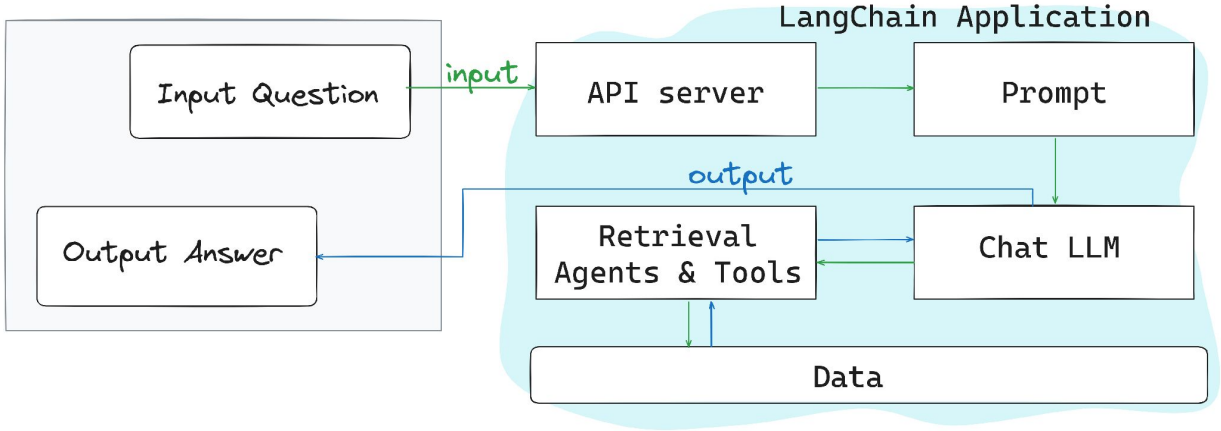
- **Contextual Understanding**: LangChain can provide a deeper understanding of the context.
- **Chaining with Tools and Application**: The system can seamlessly integrate with electronic medical records (EMR) or other tools and applications.
- **Multi-Speaker Recognition**: LangChain can better differentiate between multiple speakers (like doctor and patient), a common scenario in patient examinations.



AUTOMATION OF REPORT CREATION

Report Generation

- Research indicates that up to 24% of work hours can be devoted to administrative tasks. By leveraging generative AI, we have the opportunity to develop tools that can streamline routine tasks for doctors, enhancing efficiency.
- How great it would be, if we can automate report pre-generation?

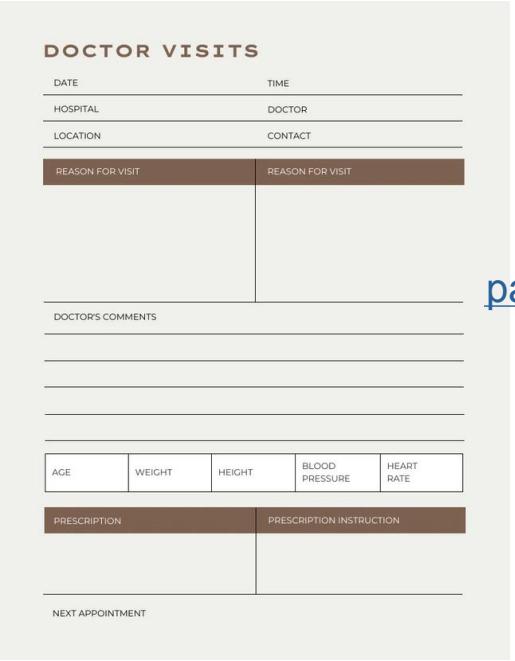


Rao, Sandhya K., et al. "The impact of administrative burden on academic physicians: results of a hospital-wide physician survey." Academic Medicine 92.2 (2017): 237-243.

AUTOMATION OF REPORT CREATION

Template Creation

- Let’s create a markdown report template:



parse

```
# Doctor Visit Report

## General Information

| Date   | Time   |
| ------ | ------ |
| {date} | {time} |

| Hospital   | Doctor   |
| ---------- | -------- |
| {hospital} | {doctor} |

...

## Next Appointment

Reason for Next Appointment:
{next_appointment_reason}
```

create

Doctor Visit Report

General Information

Date Time
- -

Hospital Doctor
- -

Name Location Contact
- - -

Reason for Visit

Reason for Visit Reason for Visit
- -

Doctor's Comment

-

Health Assessment

Age Weight Height Blood Pressure Heart Rate
- - - - -

Prescription

Prescription Prescription Instructions
- -

Next Appointment

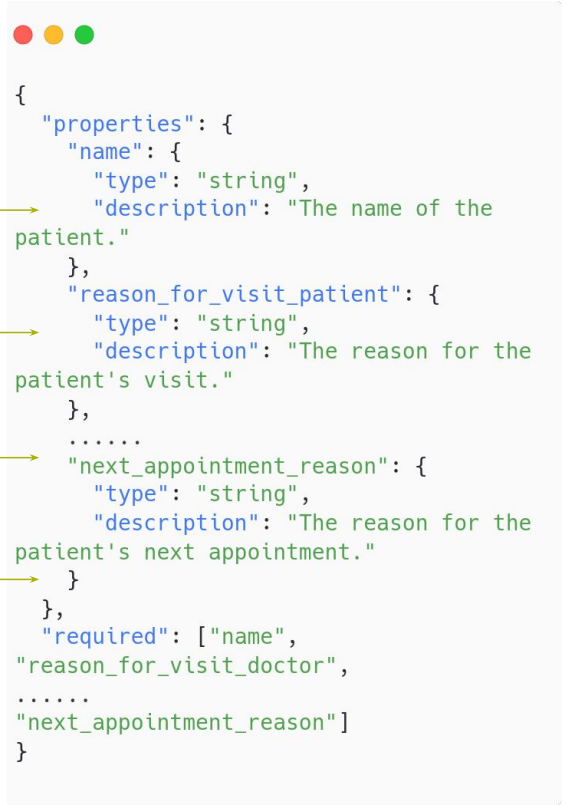
Reason for Next Appointment: -

AUTOMATION OF REPORT CREATION

Data Extraction

Below are doctors' notes:

Patient is John Baker. The patient's reason for the visit was due to cough and high temperature, which should be noted as fever and cough. Physical examination reveals no significant abnormalities in vital signs, with a blood pressure reading of 120/80 mmHg and a heart rate of 72 bpm. No visible edema or significant weight changes noted; the patient weighs 160 lbs and has a height of 5'10". The patient is in good health and prescribed antibiotics with instructions to take one tablet every 8 hours with food. Additionally, the patient's next appointment is scheduled for a follow-up in 2 weeks.



```
{
  "properties": {
    "name": {
      "type": "string",
      "description": "The name of the
patient."
    },
    "reason_for_visit_patient": {
      "type": "string",
      "description": "The reason for the
patient's visit."
    },
    ......
    "next_appointment_reason": {
      "type": "string",
      "description": "The reason for the
patient's next appointment."
    }
  },
  "required": ["name",
"reason_for_visit_doctor",
......
"next_appointment_reason"]
}
```

AUTOMATION OF REPORT CREATION

Report Generation

Now let's see, how our extracted data is used to generated a pdf report. Additional information can be collected from a database (Location, Contact) or added by system (Date, Hospital, Doctor).

```
{
 'name': 'John Baker',
 'reason_for_visit_patient': 'cough and high
temperature',
 'reason_for_visit_doctor': 'fever and cough',
 'blood_pressure': '120/80 mmHg',
 'heart_rate': '72 bpm',
 'weight': '160 lbs',
 'height': '5\'10"',
 'prescription': 'antibiotics',
 'prescription_instruction': 'take one tablet
every 8 hours with food',
 'next_appointment_reason': 'follow-up in 2 weeks'
}
```


Doctor Visit Report

General Information

Date	Time
2024-01-14	10:00 AM

Hospital	Doctor
Some Medical Center	Dr. Smith

Name	Location	Contact
John Baker	123 Main Street	555-123-4567

Reason for Visit

Reason for Visit	Reason for Visit
cough and high temperature	fever and cough

Doctor's Comment

-

Health Assessment

Age	Weight	Height	Blood Pressure	Heart Rate
35	160 lbs	5'10"	120/80 mmHg	72 bpm

Prescription

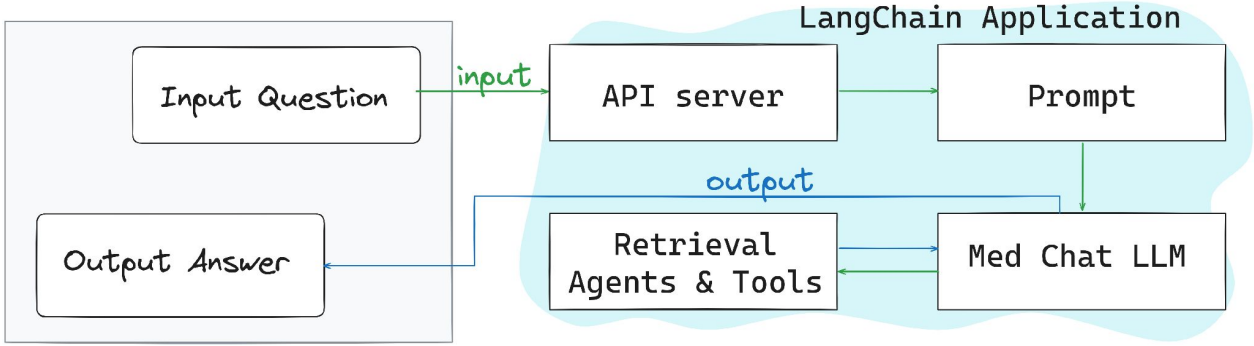
Prescription	Prescription Instructions
antibiotics	take one tablet every 8 hours with food

Next Appointment

Reason for Next Appointment: follow-up in 2 weeks

Brainstorming Assistant

LangChain can assist medical professionals by offering various differential diagnoses based on symptomatology, patient history, and epidemiological data



BRAINSTORMING ASSISTANT

Brainstorming with LangChain

Case #1

- **Patient Examination**: "A 45-year-old male presents with a persistent dry cough, shortness of breath, and intermittent chest pain over the past two weeks. Recent travel history includes a visit to a region known for high pollution levels. No significant family history of respiratory diseases."

- **Hypothesis**: There is a likelihood of an environmental lung condition, such as bronchitis or early stages of COPD, particularly considering the patient's recent exposure to high pollution. Further diagnostic tests like spirometry and a chest X-ray are recommended to confirm.

AI-generated content

Case #2

- **Patient Examination**: "An 8-year-old child presents with a high fever, sore throat, and swollen lymph nodes. The child's school recently reported a strep throat outbreak."

- **Hypothesis**: There is a high probability of strep throat, given the symptoms and contextual information about the school outbreak. A rapid strep test is recommended to confirm the diagnosis and, if positive, a course of antibiotics should be considered.

AI-generated content

FULLSTACK LANGCHAIN APPLICATION

Step 1. Speech to Data

Recording of Dr. Emily Larson, March 3, 2027:

Patient is a 48-year-old male presenting with nonspecific symptoms of fatigue and occasional mild lower back pain. On examination, his vital signs were mostly normal, though his blood pressure was a bit high at 135/85 mmHg. No visible edema or significant weight changes were noted. The patient mentions the increased frequency of nocturia but dismisses it as inconsequential. Palpation of the kidney region shows no discomfort or enlargement. Advised patient to maintain hydration and follow a balanced diet, pending lab results.



```
{
  'gender': 'male',
  'age': 48,
  'systolic_blood_pressure': 135,
  'diastolic_blood_pressure': 85,
  'symptoms': 'fatigue and occasional mild lower back discomfort',
  'notable_details': 'increased frequency of nocturia'
}
```

FULLSTACK LANGCHAIN APPLICATION

Step 2. AI-generated Hypotheses

- **Potential Illnesses**:
- 1. Hypertension: The patient's blood pressure is elevated (135/85 mmHg), which could indicate stage 1 hypertension. This could be contributing to the fatigue.
- 2. Kidney Issues: The symptoms of fatigue, mild lower back discomfort, and increased frequency of nocturia (nighttime urination) could suggest kidney problems as a kidney infection or kidney stones.
- 3. Prostate Problems: In middle-aged males, an enlarged prostate can cause symptoms like nocturia and lower back discomfort.
- 4. Diabetes: Increased urination and fatigue are common symptoms of diabetes.

- **Diagnostic Methods:**
- 1. Blood Pressure Monitoring: Regular monitoring of blood pressure to confirm hypertension.
- 2. Blood Tests: To check for kidney function (creatinine, BUN), electrolytes, blood sugar levels, and a complete blood count (CBC) to rule out anemia or infection.
- 3. Urinalysis: To check for signs of infection, kidney disease, or diabetes.
- 4. Imaging: Ultrasound or CT scan of the kidneys and urinary tract to look for stones, abnormalities, or signs of prostate enlargement.
- 5. Prostate-Specific Antigen (PSA) Test: To screen for prostate issues, including prostate cancer.

AI-generated content

FULLSTACK LANGCHAIN APPLICATION

Step 3. Examination Lab Reports

Results from March 6, 2027

Test/Examination	Result	Reference Range	Notes
Complete Blood Count (CBC)			
Hemoglobin	13.8 g/dL	13.5-17.5 g/dL	Normal
White Blood Cell Count	6,000 cells/mm³	4,500-11,000 cells/mm³	Normal
Platelets	250,000 cells/mm³	150,000-450,000 cells/mm³	Normal
Serum Creatinine (SCr)			
	1.2 mg/dL	0.6-1.2 mg/dL	Upper limit of normal
Blood Urea Nitrogen (BUN)			
	15 mg/dL	7-20 mg/dL	Normal
Urinalysis			
Protein	Trace	Negative	Slightly abnormal
Blood	Negative	Negative	Normal
Blood Pressure			
	135/85 mmHg	<120/80 mmHg	Elevated

Results from May 6, 2027

Test/Examination	Result	Reference Range	Notes
Complete Blood Count (CBC)			
Hemoglobin	12.5 g/dL	13.5-17.5 g/dL	Slightly low
White Blood Cell Count	5,800 cells/mm³	4,500-11,000 cells/mm³	Normal
Platelets	260,000 cells/mm³	150,000-450,000 cells/mm³	Normal
Serum Creatinine (SCr)			
	1.7 mg/dL	0.6-1.2 mg/dL	Abnormally high
Blood Urea Nitrogen (BUN)			
	22 mg/dL	7-20 mg/dL	Slightly high
Urinalysis			
Protein	1+	Negative	Abnormal
Blood	Trace	Negative	Slightly abnormal
Blood Pressure			
	140/90 mmHg	<120/80 mmHg	Elevated

FULLSTACK LANGCHAIN APPLICATION

Step 4. AI-generated Follow-up Hypothesis

- Hypothesis: Elevated serum creatinine, presence of protein in the blood and blood urea nitrogen levels, along with proteinuria and microscopic hematuria, suggest **kidney dysfunction**. It is important to note that initially proposed illnesses can coexist, and one condition may exacerbate another (e.g., hypertension can worsen kidney issues). The patient's slightly low hemoglobin (Hb) level may also contribute to fatigue and should be monitored.
- Suggested Treatment: Further diagnostic tests such as a renal ultrasound, referral to a nephrologist, and treatment of the underlying cause (e.g., medication, dietary changes, management of blood pressure).

AI-generated content

Test/Examination	Result	Reference Range	Notes
Complete Blood Count (CBC)			
Hemoglobin	12.5 g/dL	13.5-17.5 g/dL	Slightly low
White Blood Cell Count	5,800 cells/mm³	4,500-11,000 cells/mm³	Normal
Platelets	260,000 cells/mm³	150,000-450,000 cells/mm³	Normal
Serum Creatinine (SCr)			
	1.7 mg/dL	0.6-1.2 mg/dL	Abnormally high
Blood Urea Nitrogen (BUN)			
	22 mg/dL	7-20 mg/dL	Slightly high
Urinalysis			
Protein	1+	Negative	Abnormal
Blood	Trace	Negative	Slightly abnormal
Blood Pressure			
	140/90 mmHg	<120/80 mmHg	Elevated

FULLSTACK LANGCHAIN APPLICATION

Step 5. Clinical Decision Making

Recording of Dr. Marcus Wei, May 10, 2027:

- This is a follow-up on Mr. Smith, complaining previously on fatigue and back pain. Over the last 2 months, his kidney function has clearly declined, as there is a significant increase in creatinine from 1.2 to 1.7 mg/dL. His blood pressure is staying elevated around 140/90 despite claimed lifestyle modifications. He now reports increased fatigue, some ankle swelling and decreased urination. Looking back at the urinalysis, I've noticed there was microscopic hematuria and proteinuria overlooked before that could indicate kidney pathology. Given the progression of symptoms and new lab findings, I'm ordering a renal ultrasound to look for any structural changes or chronic kidney disease.

- In the meantime, we will monitor his diet and fluids closely. We should also start him on antihypertensives to better control the blood pressure and take some stress off the kidneys. We need to keep a close eye on Mr. Smith, given the concerning decline in kidney function over a short timeframe.

- **Diagnosis: Early Stage of Kidney Disease, Possibly Chronic.**

FULLSTACK LANGCHAIN APPLICATION

Synthetic Data for Rare Diseases

- GANs generate patient-like EHRs, labs, imaging
- Solves data scarcity in rare conditions
- Enables model training without real patients

For rare diseases, data is scarce.

Generative AI can create realistic synthetic data sets that protect privacy while enabling AI training. This is especially valuable in conditions like pediatric cancers or ultra-rare genetic diseases.



FULLSTACK LANGCHAIN APPLICATION

Medical Imaging – Image Enhancement

- GANs for low-dose CT, accelerated MRI
- Denoising, super-resolution
- Faster scans, reduced radiation exposure

Generative AI improves imaging: low-dose CT scans become high-quality, and MRI acquisition time drops from 45 minutes to under 10. GANs reconstruct full images from partial scans, improving efficiency and patient comfort.

FULLSTACK LANGCHAIN APPLICATION

Medical Imaging – Synthetic Data

- VAEs, GANs generate pathology slides
- Augment scarce training data
- Improves cancer detection AI performance

Histopathology suffers from limited labeled data.

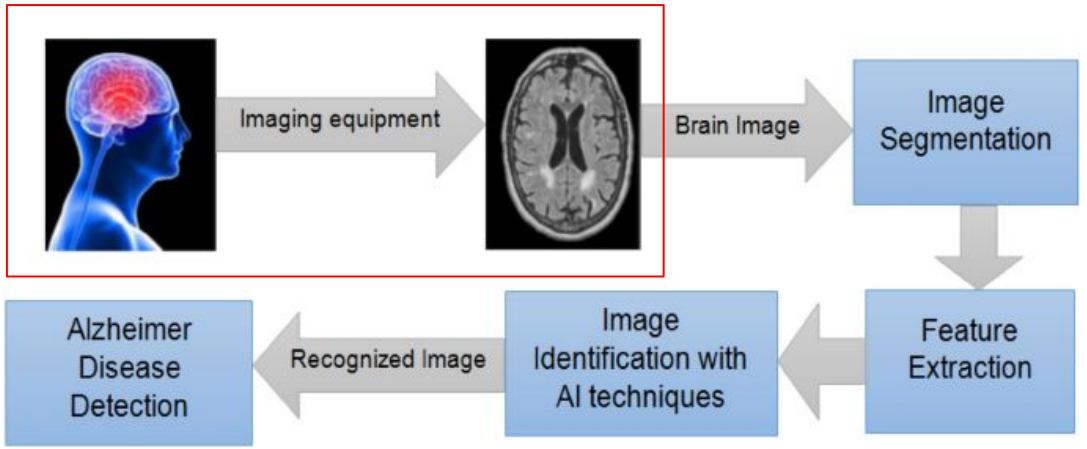
Generative AI generates synthetic slides that augment training datasets, allowing cancer-detection models to improve sensitivity and reduce false negatives.

FULLSTACK LANGCHAIN APPLICATION

Diagnostics – Automated Detection

- AI enhances radiology/pathology workflows
- GenAI generates missing modalities (e.g., PET from MRI)
- Improves anomaly detection

Generative AI is now capable of modality translation — for example, generating PET-like scans from MRI data. This aids anomaly detection and can reduce the need for multiple costly imaging sessions.



Bias & Data Quality

- AI bias from non-representative datasets
- LLM hallucinations = risk in healthcare
- Validation pipelines essential

Generative AI can hallucinate or reproduce bias. In healthcare, this is dangerous. Models must be validated rigorously, and clinicians must remain the final decision-makers.



Data Privacy & Security

- HIPAA, GDPR, Health Data Law compliance
- Patient consent & encryption
- Synthetic data as privacy solution

Privacy is paramount.

Any GenAI system must comply with HIPAA and GDPR. One promising solution is privacy-preserving synthetic data, which allows model training without exposing real patient records.

MENA Landscape

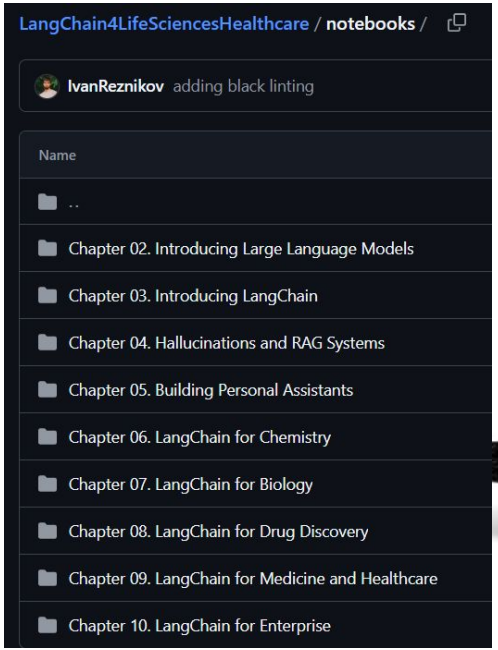
Market: $290M (2023) → $1.8B (2029) and Saudi Vision 2030: $100B AI fund

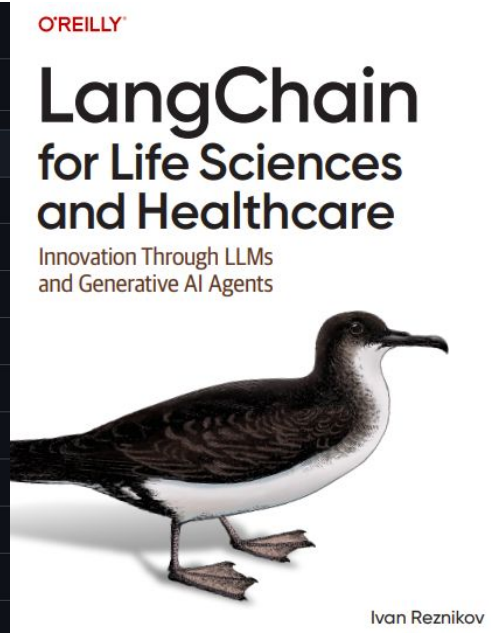
The MENA region is investing heavily. The healthcare AI market will grow six-fold by 2029. UAE leads with national genome programs and an AI minister. Saudi's Vision 2030 includes $100B AI investments.

MENA Case Studies

- Altibbi – Arabic AI triage platform
- Navina (KSA) – AI-enhanced EHR insights
- M42 (UAE) – Emirati Genome Program
- Care.ai (UAE) – Virtual nursing assistan

Altibbi, Jordan-based, runs the region's largest Arabic AI health platform. Navina in Saudi integrates AI into EHRs. M42 manages the Emirati Genome Project. Care.ai has deployed smart nursing assistants in UAE hospitals.


LangChain4LifeSciencesHealthcare / notebooks /
IvanReznikov adding black linting
Name
..
Chapter 02. Introducing Large Language Models
Chapter 03. Introducing LangChain
Chapter 04. Hallucinations and RAG Systems
Chapter 05. Building Personal Assistants
Chapter 06. LangChain for Chemistry
Chapter 07. LangChain for Biology
Chapter 08. LangChain for Drug Discovery
Chapter 09. LangChain for Medicine and Healthcare
Chapter 10. LangChain for Enterprise


O'REILLY
LangChain
for Life Sciences
and Healthcare
Innovation Through LLMs
and Generative AI Agents
Ivan Reznikov

Q&A

Now the fun part …